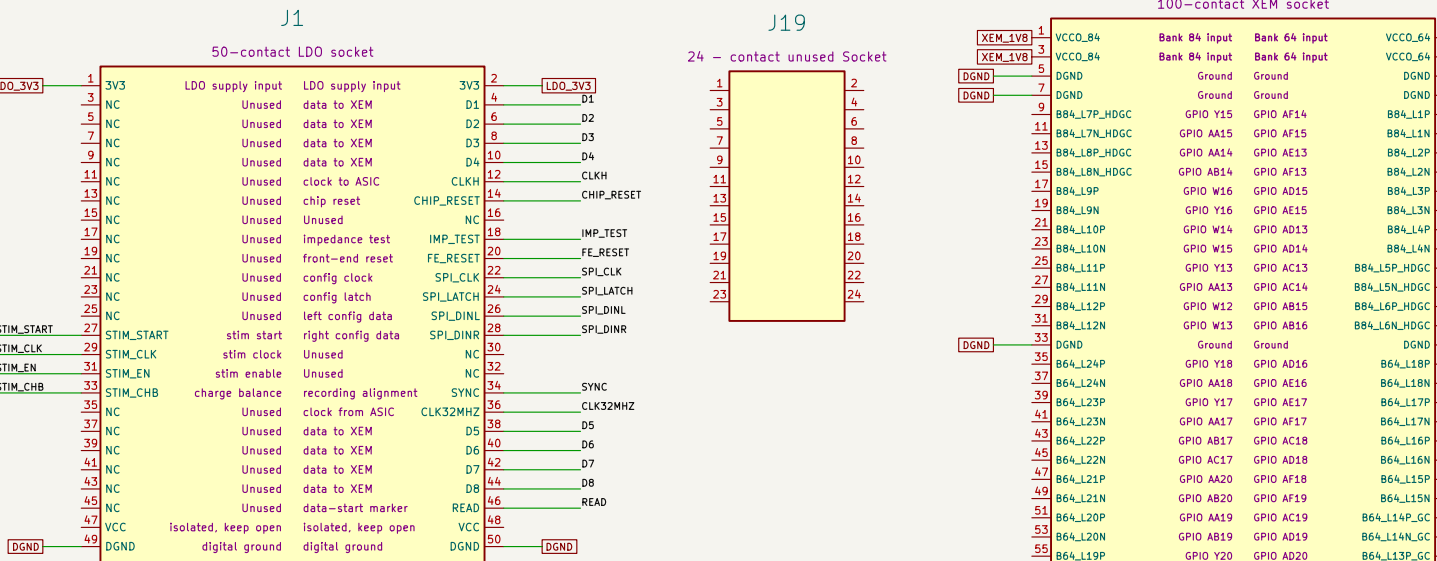
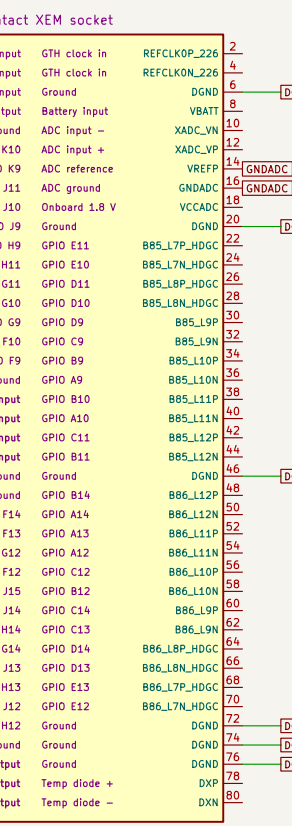


CONNECTORS – preserved foundation pin map

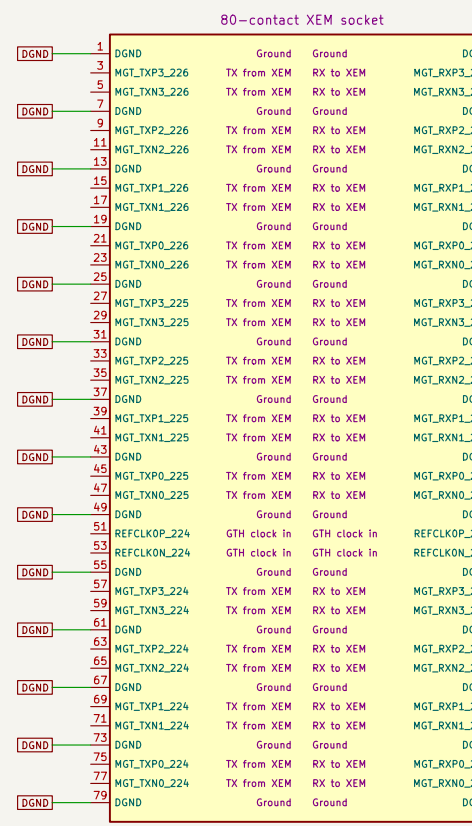
MC1



MC2



MC3



POWER FLOW

12 V enters J4 and becomes VDCIN.

VDCIN powers the XEM through MC1.

U10 / L14 make BUCK_V3; J1 connects it to LDO_V3.

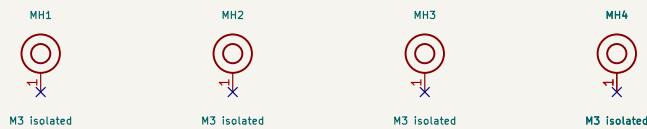
LDO_V3 feeds LDO J1 pins 1 and 2.

XEM 1.8 V directly feeds all four FPGA banks; ASIC supplies stay on the LDO board.

3 TEST / EXTERNAL 3.3 V INPUT



8 XEM MOUNTING / ISOLATED COPPER RINGS



Four plated holes 3.2 mm bore / 6.4 mm ring. Electrically isolated.

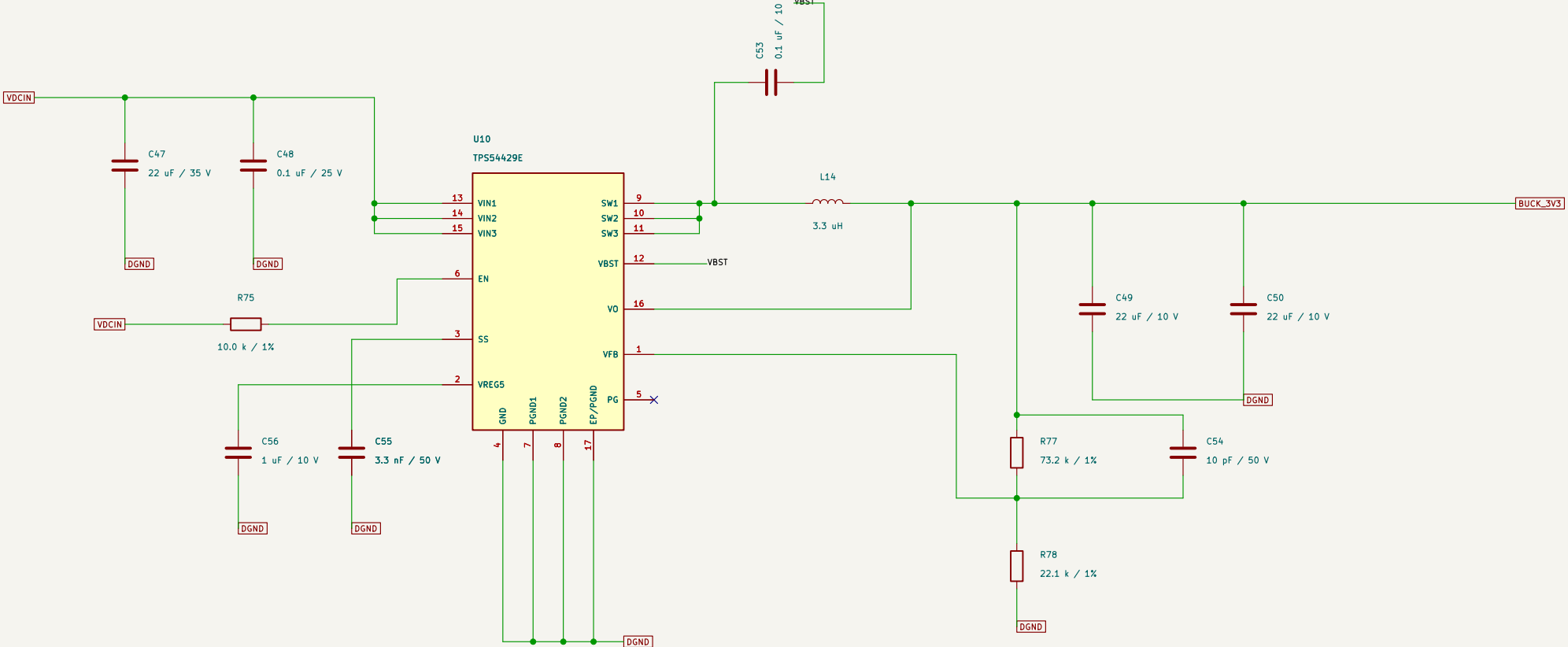
XEM nominal connector stack: 5.0 mm; choose matching standoffs.

5 INTERNAL FPGA MONITOR REFERENCE



2 3.3 V SUPPLY FOR THE EXISTING LDO BOARD

12 V -> BRK converter -> J1 -> LDO J1 pins 1 / 2



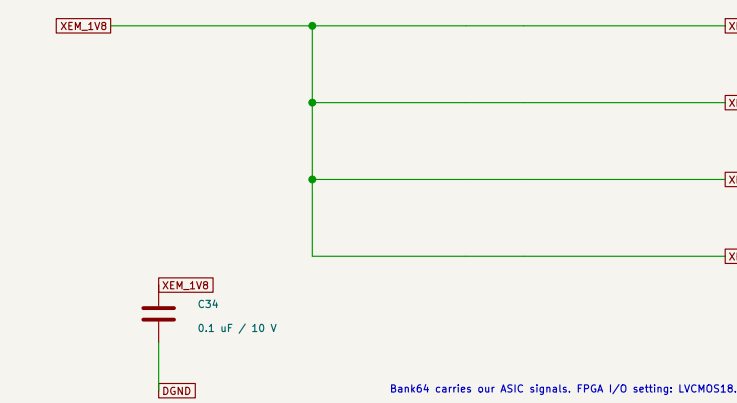
C47 / C48: input bypass. L14 + C49 / C50: output filter. R77 / R78 + C54: feedback.

C55: bootstrap. C56: soft start. C56: bias bypass. R75: always-on enable; XEM_PG line omitted.

LDO assembly revision / ground return and hardware operation remain review items.

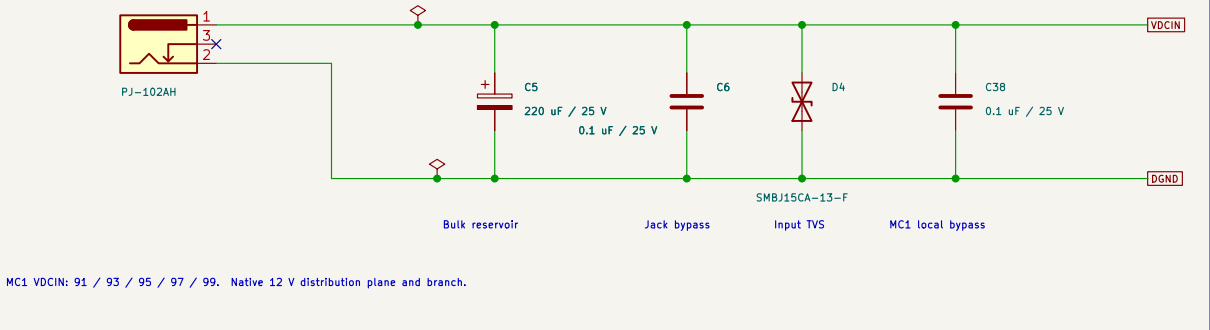
4 FPGA BANK POWER – BRK8305 A03

XEM exports 1.8 V at MC2 pins 75 / 77 / 79. No additional regulator.



1 MAIN INPUT – BRK8305 A03

External regulated 12 V – selected for BRK converter. Center positive.



6 STIMULATION POWER – LDO BOARD RESPONSIBILITY

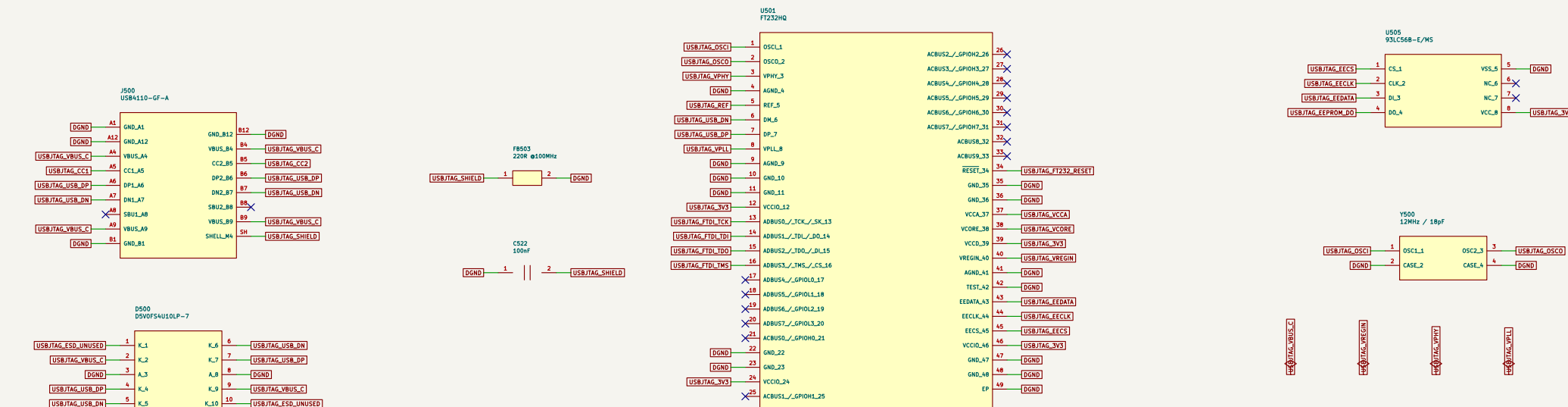
Howard relays Gerald: supply 3.3 V; the LDO board handles stimulation power.

Carrier boundary: existing LDO_V3 feed. No extra stimulation converters selected.

Older saved LDO source has separate J15 / J8 inputs; verify the matching revision.

Owner-reported boundary recorded; the older source is not proof of 3.3 V-only operation.

7 USB-C JTAG – FT232H / BRK8305-DERIVED / 1.8 V TARGET



USB VBUS powers bridge only. FT01 3.3 V is separate from LDO 3.3 V.

USB shield: BRK fitted CS22 + F8503; source DNP shields remains omitted.

Bank EEPROM requires And program; validate discovery / BCODE / UA.

53 x 83 mm carrier; isolated XEM mounts; onboard USB-C JTAG review